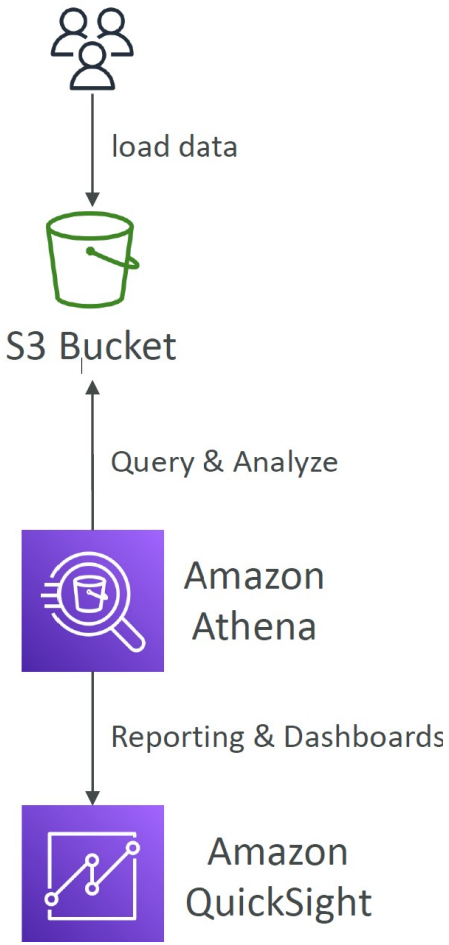
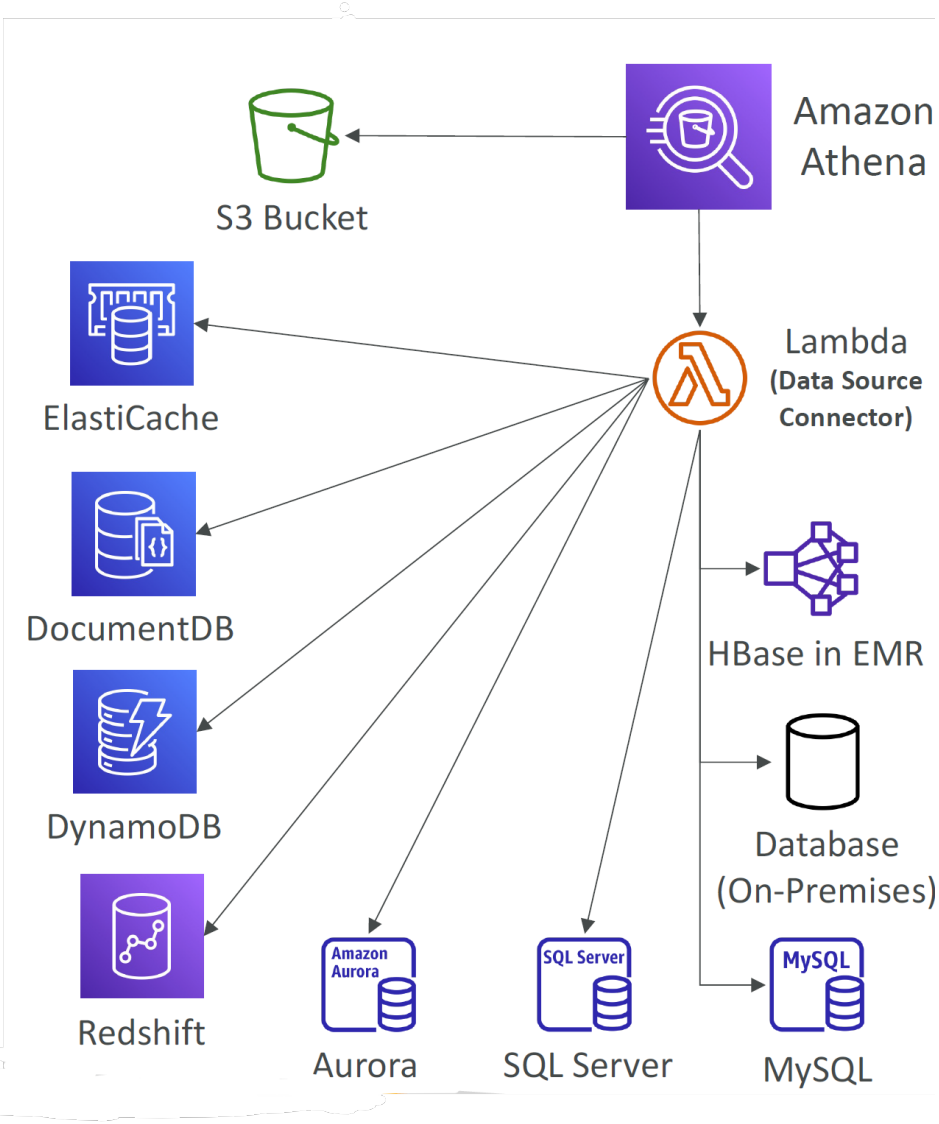
AWS Athena



- Serverless interactive query service. Lets analyze data in Amazon S3 using SQL
- Uses standard SQL language to query the files
- Supports CSV, JSON, ORC, Avro, and Parquet
- Pricing: $5.00 per TB of data scanned. Pay only per query
- Integrates with Amazon Quicksight for reporting/dashboards

AWS Athena



- Uses Data Source Connectors that run on AWS Lambda to run Federated Queries (e.g., CloudWatch Logs, DynamoDB, RDS, …)
- Store the results back in Amazon S3

For reference click here

AWS Data Exchange

• AWS Data Exchange is the only data marketplace with more than 3,500 products from over 300 providers delivered —through files, APIs, or Amazon Redshift queries— directly to the data lakes, applications, analytics, and machine learning models that use it.

• Supports financial, healthcare, geospatial, and more

• Enables secure data subscription and delivery

• Integrates with Redshift and S3

• Securely share and subscribe to third-party data within AWS.

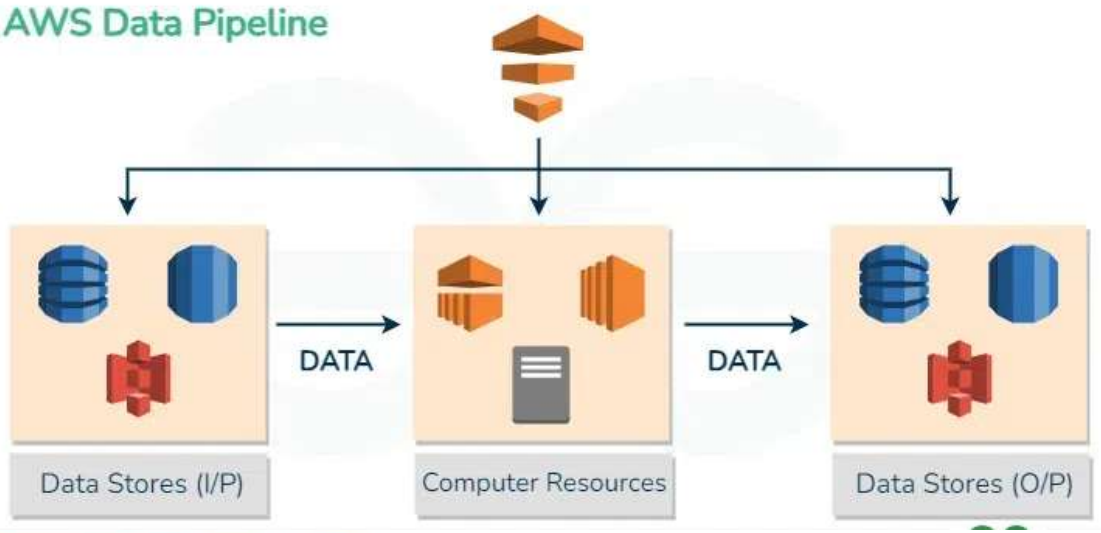
Eg: Access financial market data directly in S3.



For reference click here

AWS Data Pipeline

- Automates movement and transformation of data between compute and storage.
- ETL workflow orchestration service, Transfers data between compute & storage
- Supports scheduling, retries, dependencies
- Real-time data pipeline using IoT Core, Kinesis, Firehose, Lambda, S3, SQS, Athena, and QuickSight/Redshift for reporting.



For reference click here



Amazon EMR

Elastic Map Reduce: Creates Hadoop clusters for big data analysis and processing (hundreds of EC2 instances).

- Handles provisioning/configuration
- Autoscaling and Spot Instances integration
- Elastic and scalable clusters
- Cheaper than on-prem Hadoop
- Big data processing with Supports Hadoop, Spark, Hive, Presto.

Node Type & Purchasing: Master Node (manage cluster, long running), Core Node (run tasks, store data, long running), Task Node (optional, run tasks, usually Spot); purchasing options: On-demand, Reserved, Spot Instances.

use cases: data processing, ML, web indexing, big data.

For reference click here



AWS Glue

• Serverless ETL service (extract, transform, load) to clean, transform, and catalog data. Integrates with Lake Formation

Glue Job Bookmarks: prevent re-processing old data

Glue Elastic Views: Combines and replicates data across multiple data stores using SQL.

Glue DataBrew: clean and normalize data using pre-built transformation

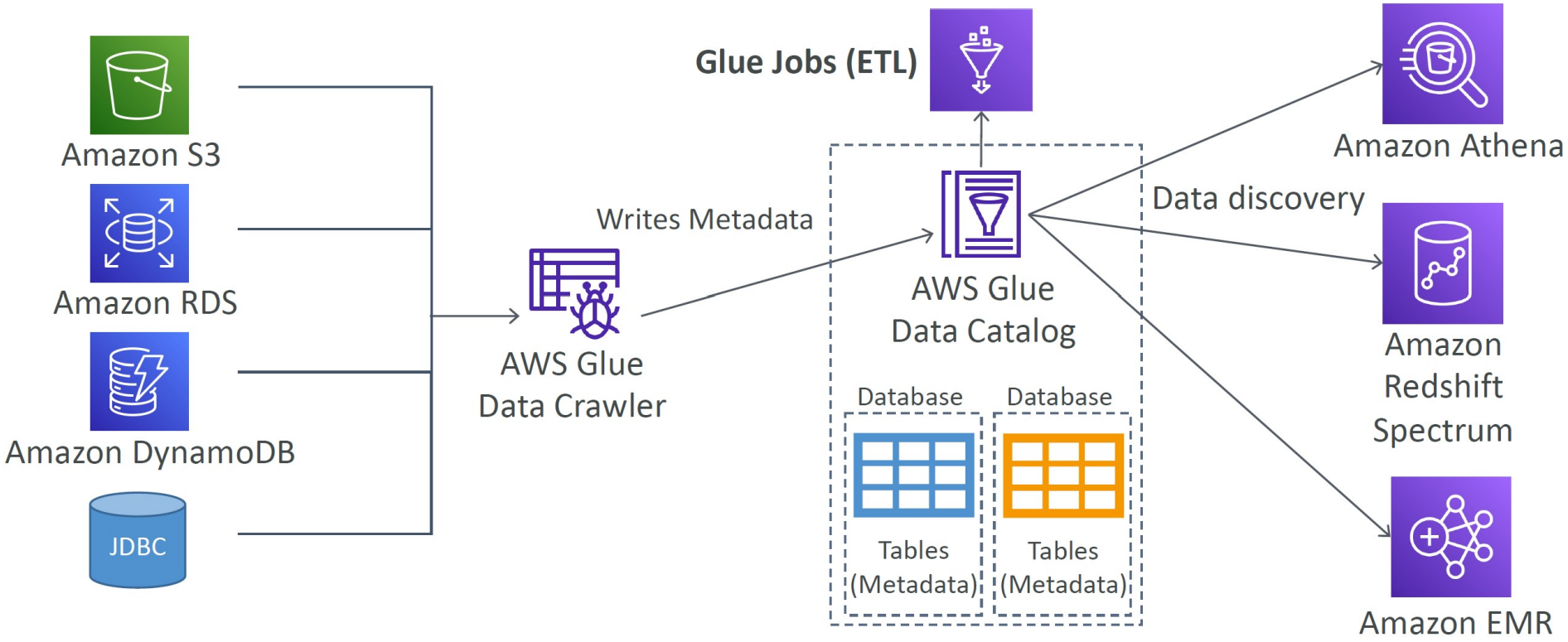
Glue Studio: new GUI to create, run and monitor ETL jobs in Glue

Glue Streaming ETL (built on Apache Spark Structured Streaming) compatible with Kinesis Data Streaming, Kafka, MSK (managed Kafka)

For reference click here

- **Glue Data Catalog:** Catalog of datasets.

Example: Prepare data for analytics in Redshift.

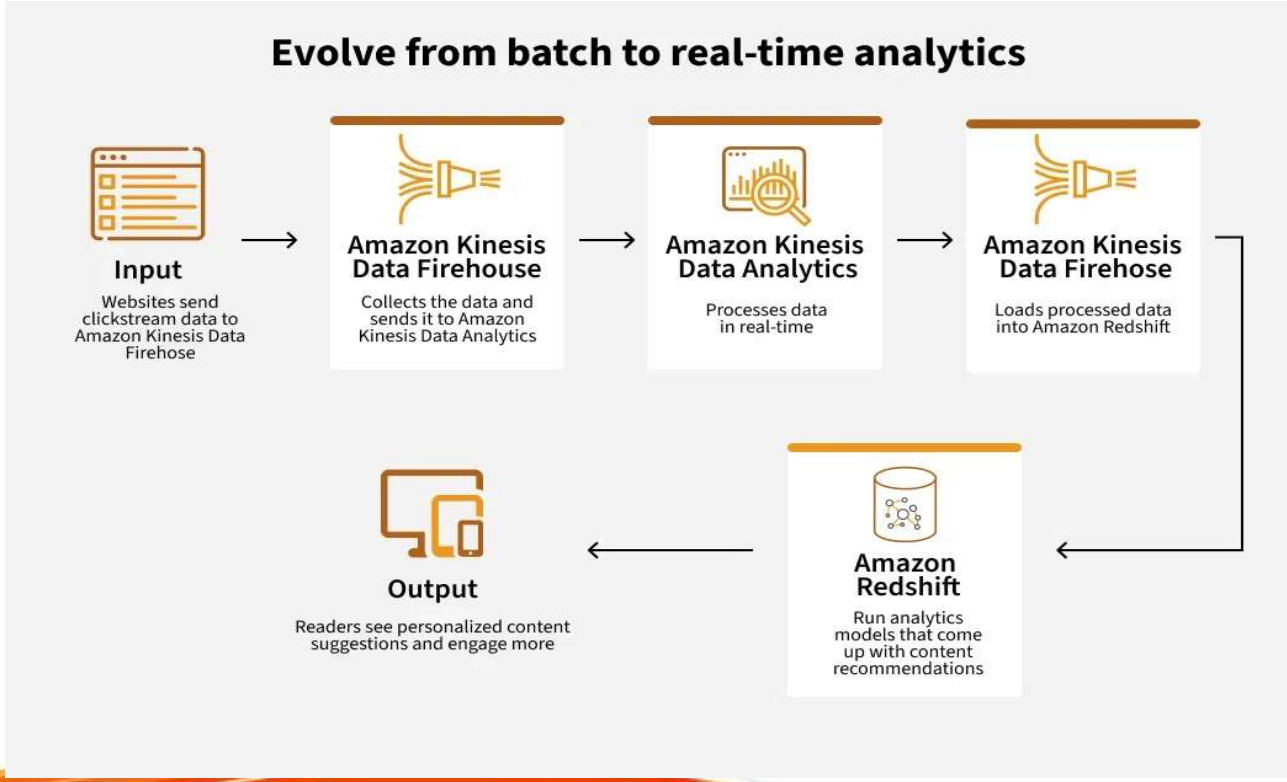


For reference click here

Amazon Kinesis

- Kinesis is great for Real-time data streaming, analytics and ETL.
- Standard: pull data - 2 MB per shard, Enhanced-fan out: push data - 2 MB per shard per consumer
- Provisioned mode or ondemand capacity mode Possibility to replay data

Eg: Analyze live IoT sensor data.



For reference click here



AWS Lake Formation

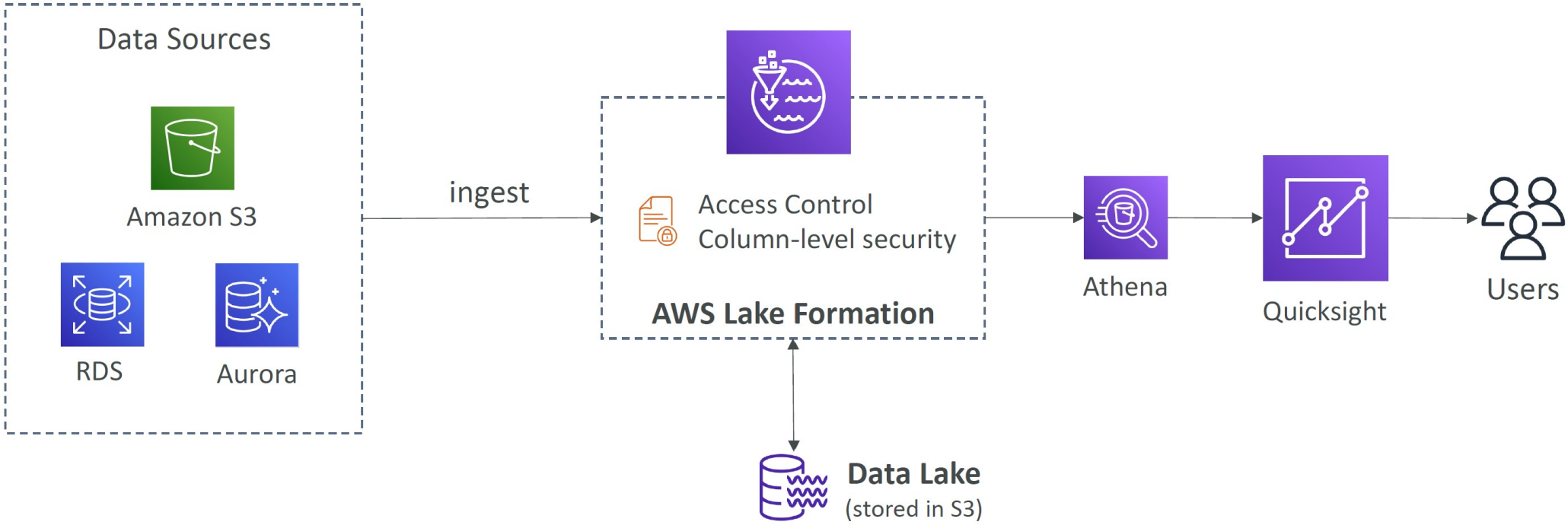
- Simplifies building secure data lakes.
- Central place to have all your data for analytics purposes
- Fully managed service that makes it easy to setup a data lake in days
- Discover, cleanse, transform, and ingest data into your Data Lake
- It automates many complex manual steps (collecting, cleansing, moving, cataloging data, …) and de-duplicate (using ML Transforms)
- Combine structured and unstructured data in the data lake
- Out-of-the-box source blueprints: S3, RDS, Relational & NoSQL DB…
- Fine-grained Access Control for your applications (row and column-level)
- Built on top of AWS Glue

Example: Centralize and govern organizational data.

For reference click here

AWS Lake Formation

Example: Centralize permission organizational data .





Amazon MSK

Fully Managed streaming platform for Apache Kafka. Alternative to Amazon Kinesis. Allow you to create, update, delete clusters.

- MSK creates & manages Kafka brokers nodes & Zookeeper nodes for you
- Deploy the MSK cluster in your VPC, multi-AZ (up to 3 for HA)
- Automatic recovery from common Apache Kafka failures
- Data is stored on EBS volumes for as long as you want
- Automatic patching and scaling, High availability
- Works with Lambda & EC2 consumers

Example: Process website clickstream events.

For reference click here



Amazon OpenSearch Service

- Successor of Amazon ElasticSearch.
- Search, monitor, and analyze data in real time.
- Managed or serverless cluster modes
- Ingests from Kinesis Data Firehose, Amazon IoT, CloudWatch Logs
- Security via Cognito, IAM, KMS, TLS.
- Includes OpenSearch Dashboard for visualization.

Example: Log analytics and monitoring.

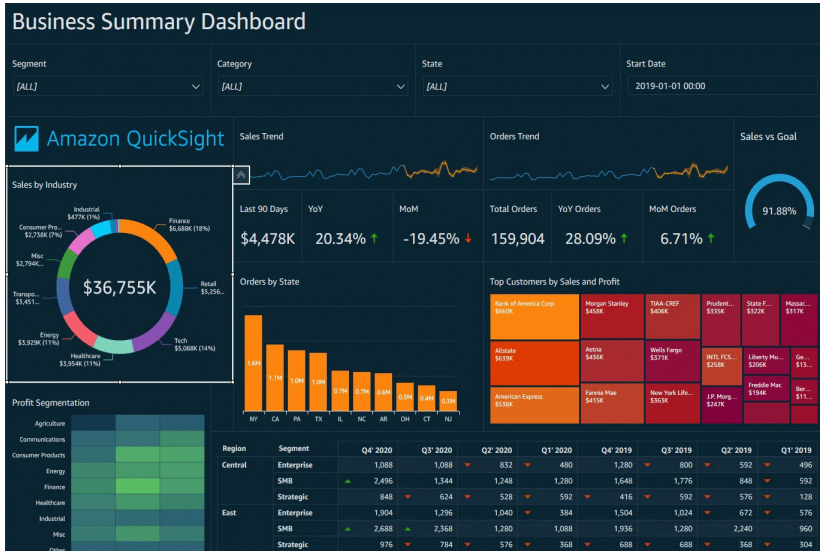
For reference click here

Amazon QuickSight

- Serverless machine learning-powered Business intelligence (BI) for interactive dashboards and reports.
- Fast, auto-scalable, embeddable, per-session pricing.
- Integrates with RDS, Aurora, Athena, Redshift, S3. Example: Visualize sales data trends.

Use cases:

- upto 300% increase in BI analytics usage
- Building visualizations
- Perform ad-hoc analysis
- Get business insights using data



For reference click here



Amazon Redshift

Based on PostgreSQL but not for OLTP. 10x better performance than other data warehouses, scales to PBs. columnar storage, parallel query engine. SQL interface, integrates with QuickSight.

- Redshift Cluster: Leader Node (query planning, results aggregation) and Compute Node (performing queries, sending results).
- Snapshots & DR: Stored incrementally in S3; can restore to a new cluster; can automatically copy snapshots to another region.
- Redshift Spectrum: Queries data in S3 without loading it, requiring a Redshift cluster to start queries.

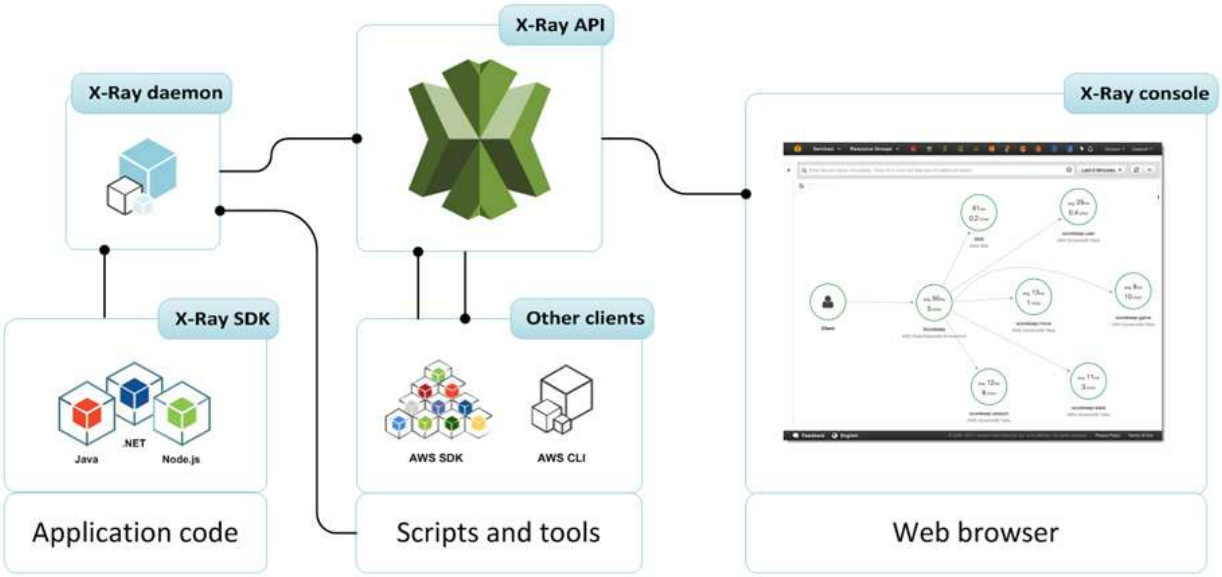
Example: Run SQL queries on large datasets.

For reference click here

AWS X-Ray

- Tracing service to analyze and debug distributed applications.
- Distributed tracing for microservices
- Debug latency & request flows. Helps identify bottlenecks
- Visualizes end-to-end paths

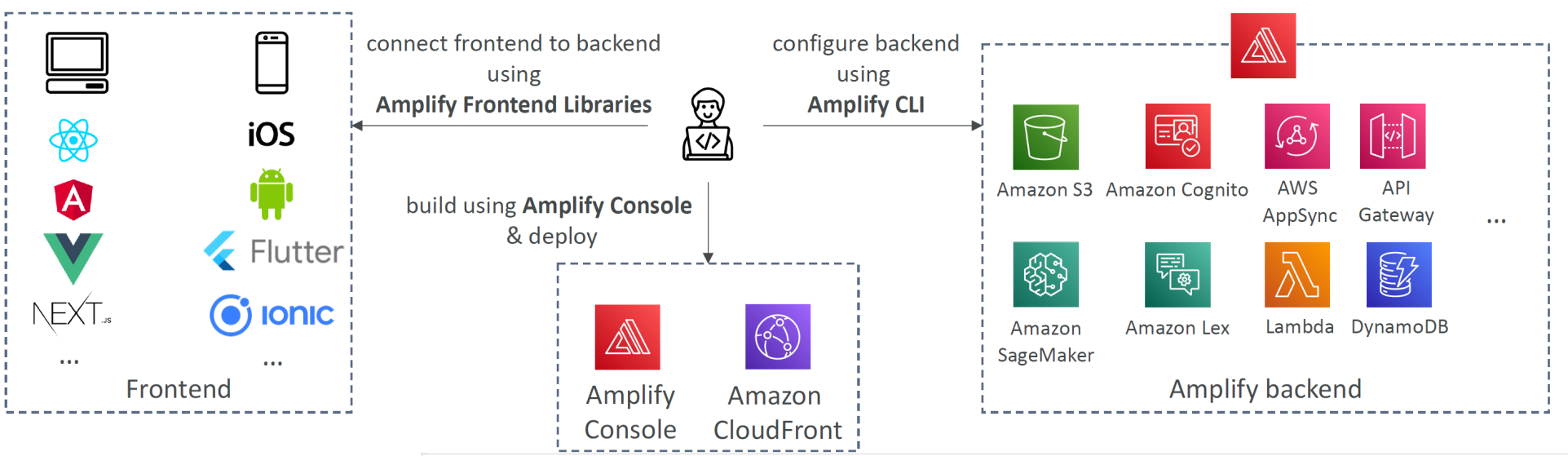
Example: Identify latency in microservices.



For reference click here

AWS Amplify

- Develop and host full-stack web and mobile apps.
- Authentication, Storage, API (REST, GraphQL), CI/CD, PubSub, Analytics, AI/ML Predictions, Monitoring
- Connect your source code from GitHub, AWS CodeCommit, Bitbucket, GitLab, or upload directly.



For reference click here



Amazon API Gateway

Create, publish, and manage secure APIs. supports WebSocket protocol. Handles API versioning and environments (dev, test, prod). Transforms/ validates requests/responses. generates SDK/API spec.

High Level Integrations: Lambda Function (expose REST API), HTTP (expose internal/external HTTP endpoints, ALB), AWS Services (expose any AWS API).

Endpoint Types:

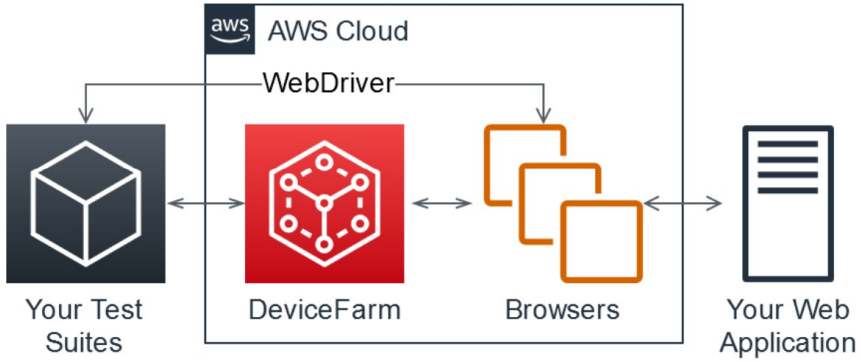
- Edge Optimized : For global clients; requests routed through CloudFront Edge locations to improve latency; API Gateway still in one region.
- Regional: For clients within the same region; can manually combine with CloudFront for more control.
- Private: Accessible only from your VPC using an interface VPC endpoint.
- Security: User Authentication via IAM Roles (internal apps), Cognito (external/mobile users), Custom Authorizer (custom logic); HTTPS security via ACM for custom domain names.

For reference click here

AWS Device Farm

- Test mobile and web apps on real devices.
- Every browser the test is executed on, Device Farm generates video recordings and Selenium logs to help you quickly identify any issues with your web app.

Example: Test Android and iOS apps remotely.

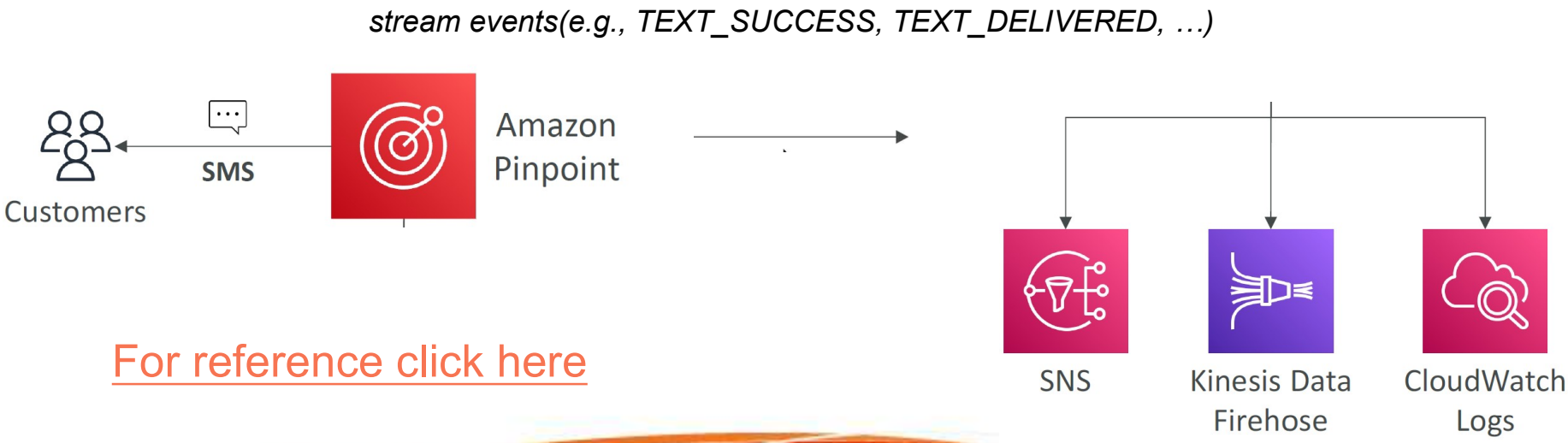


For reference click here

Amazon Pinpoint

- Engage customers with targeted messaging.
- Supports email, SMS, push, voice, and in-app messaging
- Ability to segment and personalize messages with the right content to customers
- Scales to billions of messages per day

Example: Send promotional SMS or email campaigns.



For reference click here



Amazon Comprehend

- NLP service that extracts insights from text (sentiment, entities, topics).
- Extracts key phrases, places, people, brands, or events
- Understands how positive or negative the text is
- Analyzes text using tokenization and parts of speech
- Automatically organizes a collection of text files by topic

use cases:

• analyze customer interactions (emails) to find what leads to a positive or negative experience

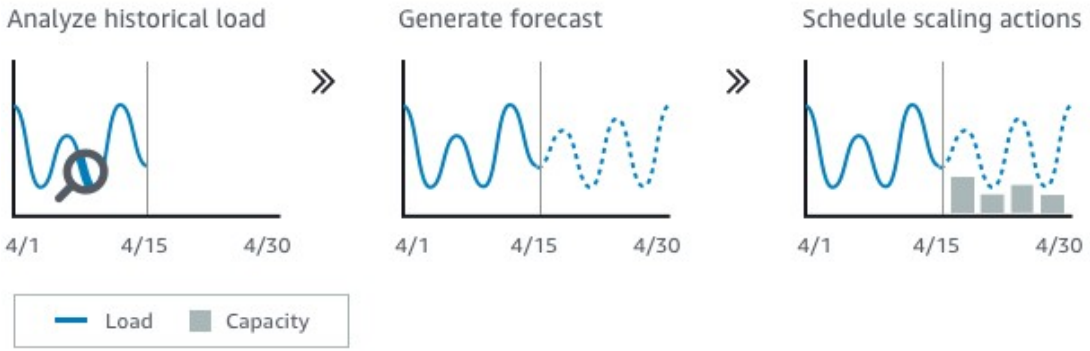
• Create and groups articles by topics that Comprehend will uncover

For reference click here

Amazon Forecast

- Machine-learning forecasting service for time-series predictions and built for business metrics analysis
- Reduce waste, increase inventory turns, and improve in-stock availability by forecasting product demand at specific probability levels
- Improve capital utilization and make long-term decisions with more confidence.

Example: Predict product demand.

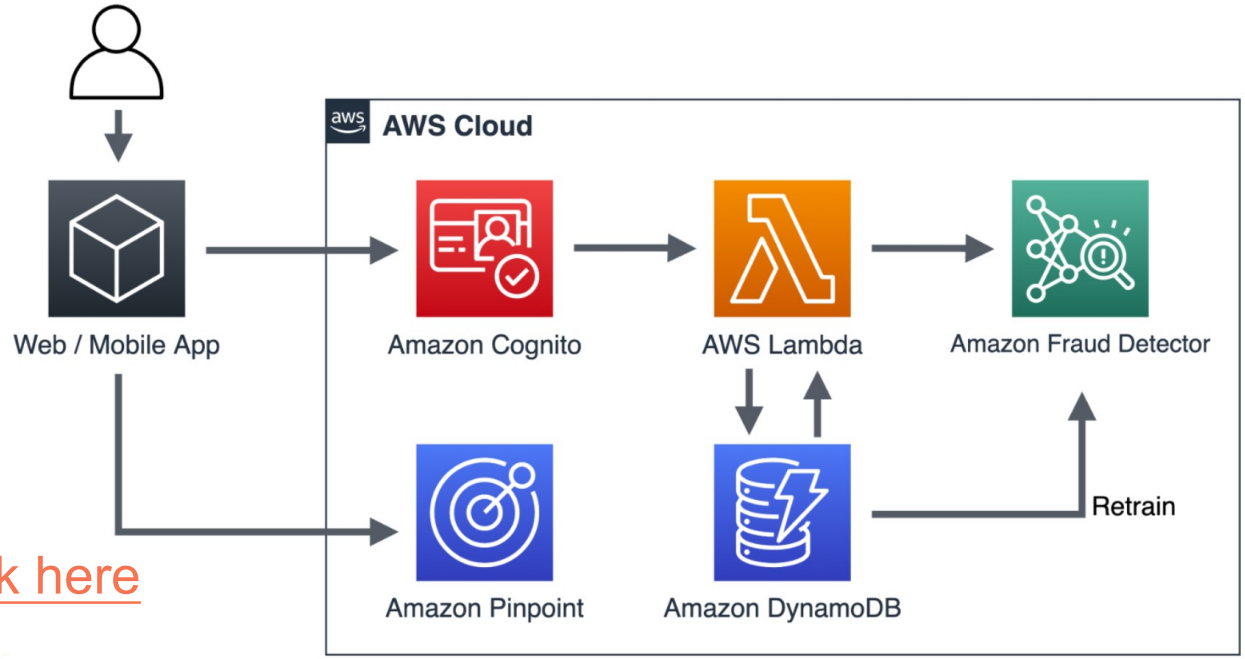


For reference click here

Amazon Fraud Detector

- Detect fraudulent online activity using ML.
- Low fraud risk – For a model score equal to or less than 650
- Medium fraud risk – For a model score between 650 and 850
- High fraud risk – For a model score over 850

Example: Training a call center fraud detection model for IVR calls, Prevent fake account sign-ups in real time with AI using Amazon Fraud Detector

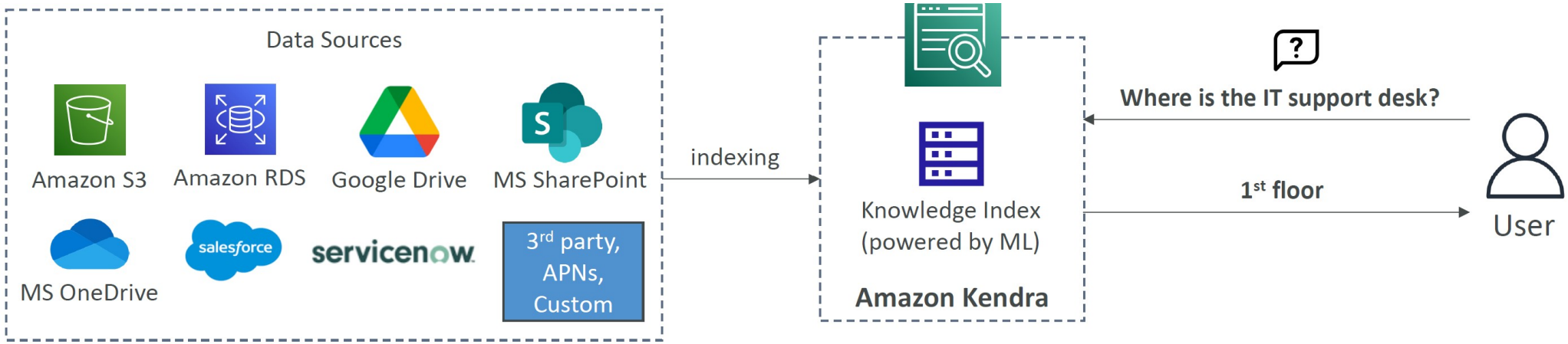


For reference click here

Amazon Kendra

- Fully managed document search service powered by Machine Learning
- Extract answers from within a document (text, pdf, HTML, PowerPoint, FAQs…)
- Intelligent enterprise search powered by AI.

Example: Search across internal documents.



For reference click here

Amazon Lex



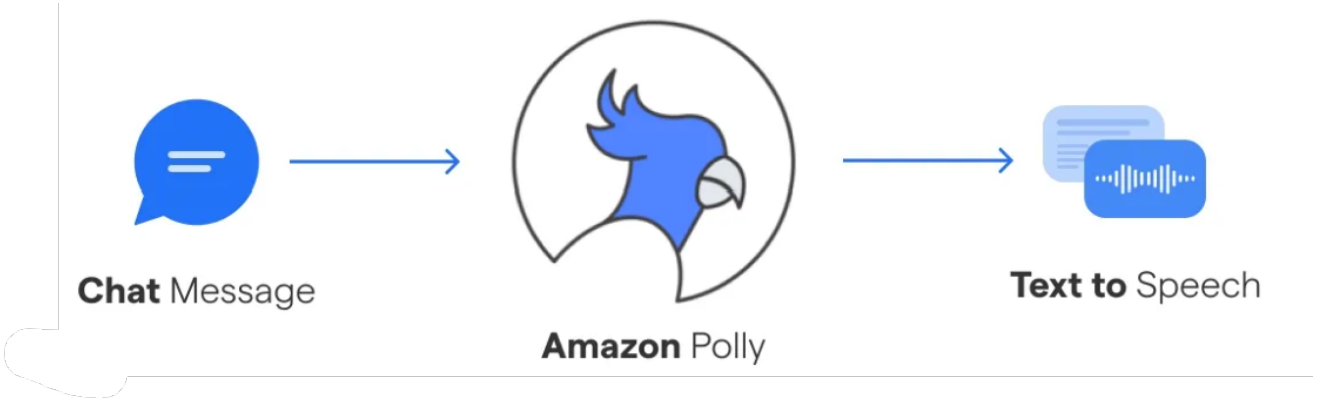
Core Features	• Automatic Speech Recognition (ASR) • Natural Language Understanding (NLU) • Multi-language support • Intents, slots, and utterances
Bot Design and Customization	• Dialog management • Context management • Lambda integration • Slot validation and prompts
Development and Deployment	• Multi-channel support • Versioning and aliases • Built-in and custom slot types • Visual conversation builder (V2 Console)
Monitoring and Analytics	• Logging and monitoring • Conversation logs • Sentiment analysis (via Amazon Comprehend)
Security and Compliance	• Data privacy and encryption • IAM integration • HIPAA eligibility

For reference click here

Amazon Polly

- Text-to-speech service that generates lifelike speech.
- Turn text into lifelike speech using deep learning

Example: Voice narration for news articles.

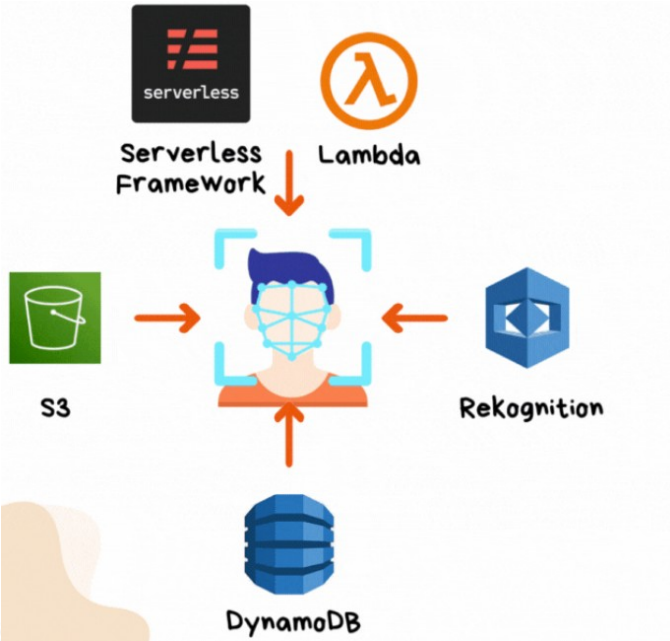


For reference click here

Amazon Rekognition

- Image and video analysis using AI. (face detection, labels, moderation).
- Find objects, people, text, scenes in images and videos using ML
- Facial analysis and facial search to do user verification, people counting
- Create a database of “familiar faces” or compare against celebrities

Example: Detect objects and faces in images.



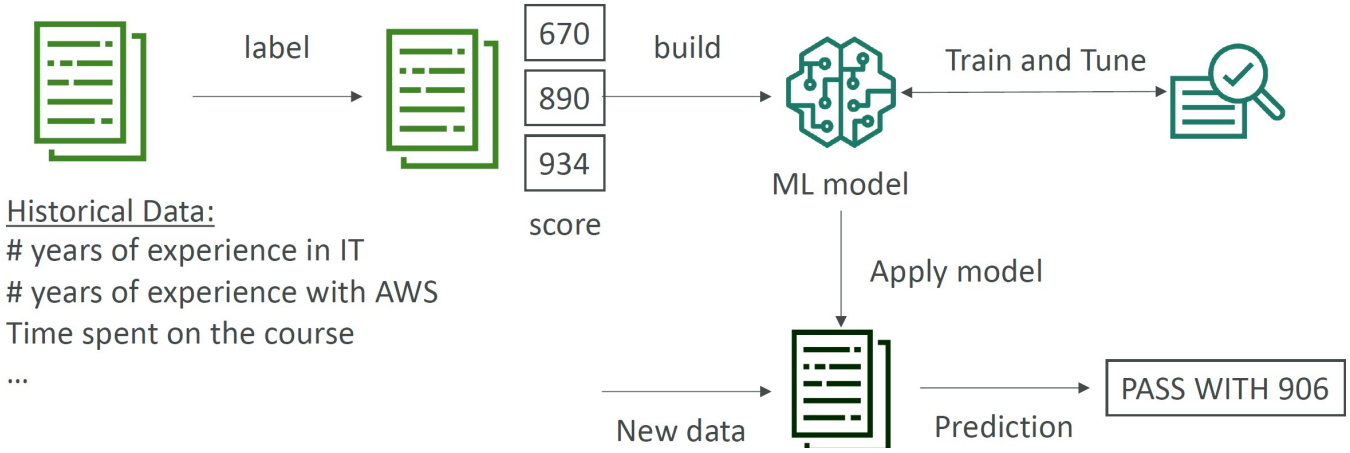
For reference click here

Amazon SageMaker

Full machine-learning platform to build, train, and deploy ML models.

Typically, difficult to do all the processes in one place + provision servers

Example: Train churn prediction model, predicting your exam score.

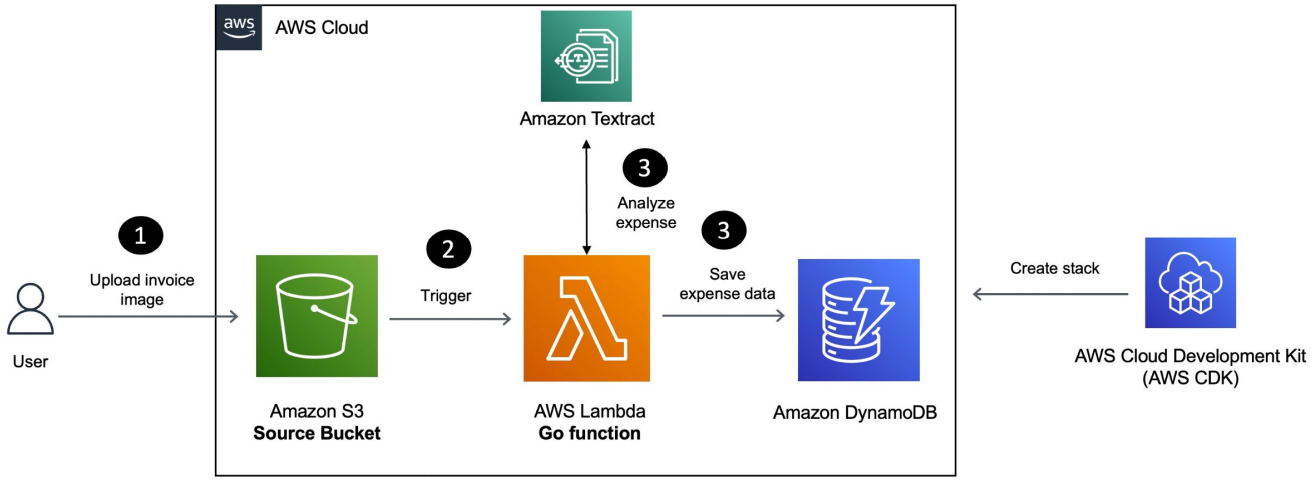


For reference click here

Amazon Textract

- Extract text, tables, and forms from scanned documents.
- Automatically extracts text, handwriting, and data from any scanned documents using AI and ML
- Extract data from forms and tables
- Read and process any type of document (PDFs, images, …)

Example: Read invoices automatically.



For reference click here

Amazon Transcribe

- Automatic convert speech-to-text service.
- Uses a deep learning process called automatic speech recognition (ASR) to convert speech to text quickly and accurately
- Automatically remove Personally Identifiable Information (PII) using Redaction
- Supports Automatic Language Identification for multi-lingual audio
- Use cases:
- transcribe customer service calls
- automate closed captioning and subtitling
- generate metadata for media assets to create a fully searchable archive



”Hello my name is Kowsalya.

For reference click here

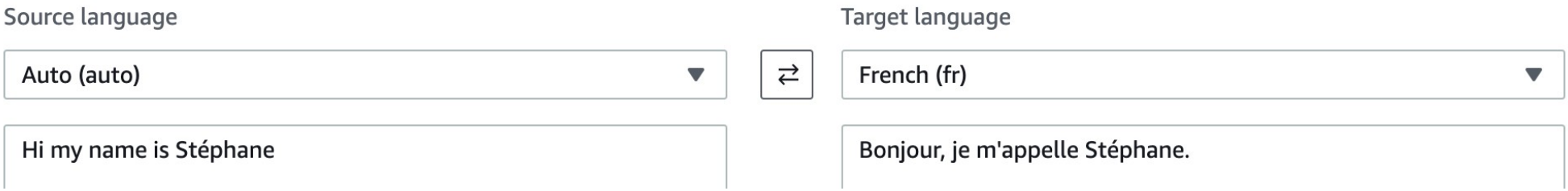
Amazon Translate

- Natural and accurate language translation
- Amazon Translate allows you to localize content - such as websites and

applications - for international users, and to easily translate large

volumes of text efficiently. Translate text between languages.

Example: Localize website content.



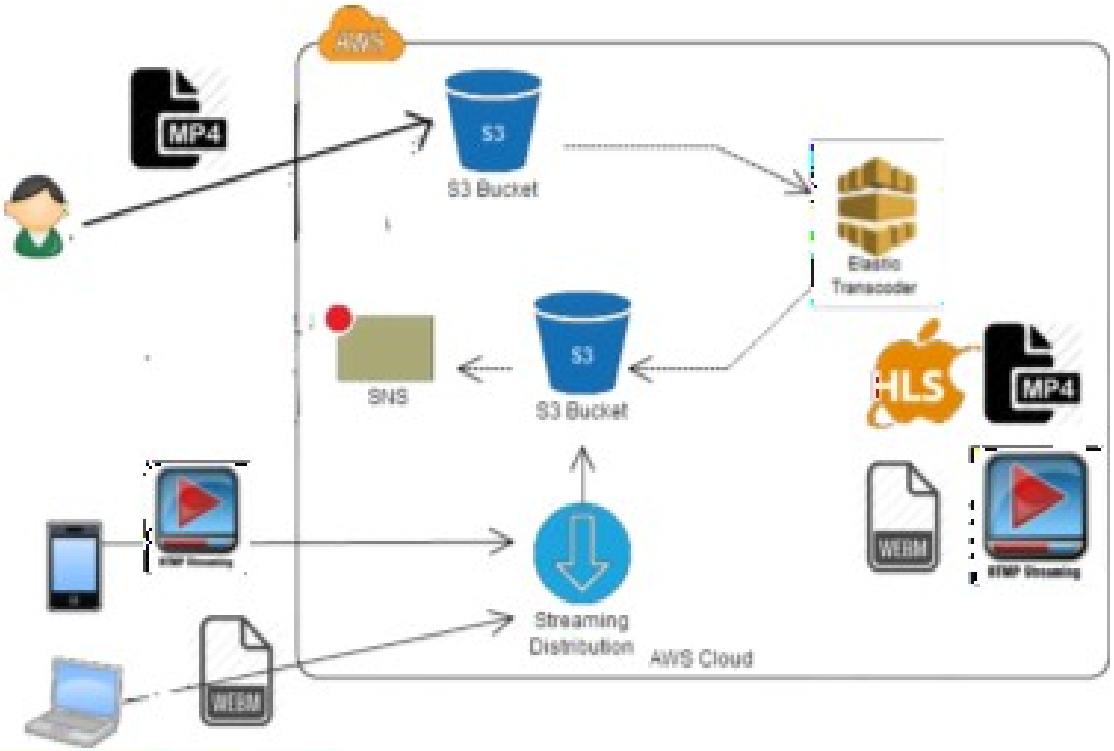
For reference click here

Amazon Elastic Transcoder

- Convert media files to multiple formats. A media transcoding service in the cloud.
- It converts video and audio files from one format to another, making them playable on different devices (mobile, web, TV).

Example: Generate mobile-friendly video formats.

- Fully managed
- Automatically scales
- Supports presets for different resolutions.



For reference click here

Amazon Kinesis Video Streams

- Amazon Kinesis Video Streams makes it easy to securely stream video from IoT (Internet of Things) connected devices to AWS for analytics, machine learning (ML), playback, and other processing
- Captures, processes, stores video Stream and analyze live video securely.

Example: Process CCTV camera feeds in real time.



For reference click here